

Detecting Tennessee Eastman faults from normal operation only

06-763 miniproject, team banana_bread_matcha_latte. PCA keeps $k = 31$ components (90.2% of the training variance). Thresholds are the 99th percentile of validation runs 301-400.

1. The plant and the task

The Tennessee Eastman Process is a simulated chemical plant with a reactor, condenser, vapour-liquid separator, recycle compressor and product stripper. Fifty-two signals are recorded every 3 minutes: 41 measurements (x_{meas_1} to x_{meas_41}) and 11 manipulated variables (x_{mv_1} to x_{mv_11}). A run is 500 samples, or 25 hours. In a faulty run the fault starts after sample 20, so samples 1-20 are normal and samples 21-500 are faulty.

The task is to detect 20 fault types with models trained on normal operation only, because a classifier trained on labelled faults cannot recognise a fault type it has not seen. We build two detectors: a PCA monitor (T2 and SPE) and a ridge forecast-residual monitor. Both are fitted on fault-free runs 1-300. Thresholds are set on fault-free runs 301-400 and false alarms are measured on fault-free runs 401-500. Faults 1-20, 20 runs each, are scored afterwards. No faulty sample is used to fit a model or set a threshold.

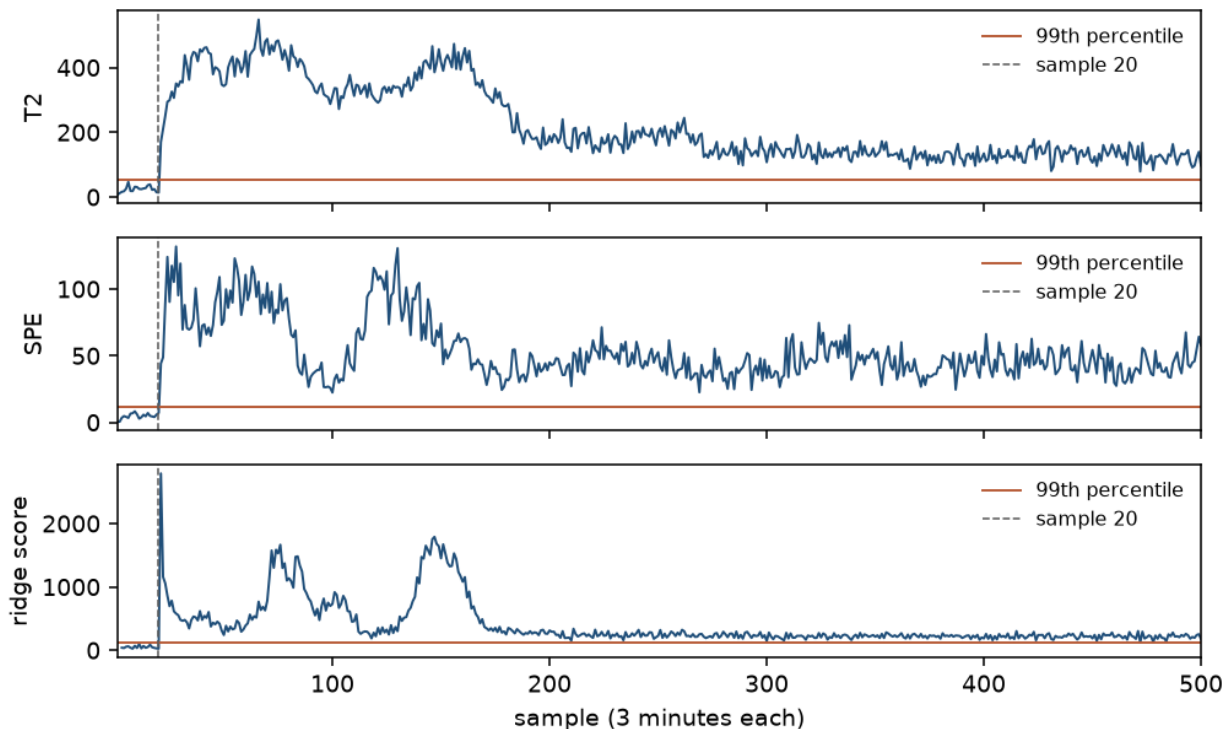
2. The two detectors

PCA monitor. Each channel is standardized with the training mean and standard deviation ($ddof = 1$). PCA keeps the smallest number of components whose eigenvalues sum to at least 90% of the total: $k = 31$ (90.2%). With loadings P and scores $t = P'z$, $T2$ is the sum of t_i^2 / λ_i over the retained components. It is large when a sample moves an unusual distance along the normal directions of variation. SPE is the squared reconstruction error $|z - PP'z|^2$. It is large when the channels stop following their normal correlations.

Forecast monitor. This is the Lecture 8 one-step residual detector applied to all 52 channels. The features are the standardized samples at $t-1$ and $t-2$ (104 columns) and the target is the sample at t . Lags never cross a run boundary, so samples 1 and 2 of each run have no score. One Ridge($\alpha = 1$) is fitted on the training runs to predict all 52 channels. Each residual is divided by that channel's training residual standard deviation ($ddof = 1$), and the score is the sum of squares. It is large when the last two samples do not predict the current one.

Thresholds and alarms. Each threshold is `numpy.quantile` of the validation scores at 0.99 with the default interpolation: $T2$ 51.8, SPE 11.77, ridge 112.4. A sample is in alarm when it and the two preceding samples of the same run all exceed the threshold. Figure 1 shows fault 7, run 1 (C header pressure loss, reduced availability (stream 4); step). The share of samples above the threshold in samples 21-40, then 200-500, is $T2$ 1.00 then 1.00, SPE 1.00 then 1.00, and ridge 1.00 then 1.00. The dashed line marks sample 20 and the horizontal line is the threshold.

Fault 7, run 1: C header pressure loss, reduced availability (stream 4); step



3. Results

Table 1 gives the detection rate and median delay for each fault and statistic. The detection rate is the share of samples after sample 20 in alarm, averaged over the 20 runs. The delay is the median, over runs that alarmed, of the minutes from sample 20 to the first alarm. "none" means all 20 runs were missed, and "N missed" counts runs with no alarm. Fault 0 is the false-alarm rate, the share of samples in alarm on test runs 401-500.

Table 1. Detection rate and median delay per fault and statistic. Fault 0 is the false-alarm rate on fault-free test runs 401-500.

Fault	T2 rate	T2 delay	SPE rate	SPE delay	Ridge rate	Ridge delay
0	0.00052	-	0	-	0	-
1	0.985	25.5 min	0.992	15 min	0.994	12 min
2	0.973	39 min	0.959	57 min	0.965	51 min
3	0.000	231 min, 19 missed	0.000	none	0.000	none
4	0.091	64.5 min	0.994	12 min	0.969	9 min
5	0.332	9 min	0.125	15 min	0.996	9 min
6	0.979	33 min	0.996	9 min	0.996	9 min
7	0.996	9 min	0.996	9 min	0.996	9 min
8	0.942	70.5 min	0.822	67.5 min	0.954	52.5 min
9	0.000	231 min, 19 missed	0.000	none	0.000	none
10	0.085	471 min	0.177	177 min	0.698	67.5 min
11	0.258	43.5 min	0.434	40.5 min	0.454	33 min
12	0.944	37.5 min	0.867	48 min	0.957	30 min
13	0.884	159 min	0.893	136.5 min	0.902	138 min
14	0.978	12 min	0.784	13.5 min	0.995	9 min
15	0.000	231 min, 19 missed	0.000	none	0.000	none
16	0.019	690 min, 3 missed	0.101	156 min	0.712	45 min
17	0.666	138 min	0.868	118.5 min	0.867	118.5 min
18	0.871	175.5 min	0.884	151.5 min	0.893	141 min
19	0.010	471 min	0.003	610.5 min, 6 missed	0.029	121.5 min
20	0.136	186 min	0.432	165 min	0.635	151.5 min

False alarms. On the test runs, T2 exceeds its threshold on 546 of 50,000 samples, SPE on 504 of 50,000 and ridge on 496 of 49,800, close to the 1% a 99th percentile implies. After the three-sample rule, 26, 0 and 0 samples are in alarm. SPE and ridge exceedances are isolated. T2 exceedances cluster, because slow normal variation lies inside the PCA subspace and keeps T2 high for several samples.

Detection. All three statistics reach a rate of 0.50 or more on faults 1, 2, 6, 7, 8, 12, 13, 14, 17, 18, and all three stay below 0.10 on faults 3, 9, 15, 19. Ridge has the highest rate, or ties for it, on 13 of 20 faults. The shortest median delay is 9 minutes, because the first three-sample window entirely after the fault ends at sample 23. An earlier alarm needs a pre-fault sample above the threshold.

A delay means at least one run alarmed, not that the fault was detected. T2 on fault 16 has a median delay of 690 minutes at a rate of 0.019, and fault 19 has delays on all three statistics at rates of 0.029 or less. T2's only alarms on faults 3, 9 and 15 start in run 7 at sample 97, in all three faults. The three T2 traces are nearly identical around that sample, which suggests that runs with the same number share their background noise. The alarm is a normal excursion, not a detection.

4. Comparison

T2 responds to large moves along normal directions, SPE to broken correlations between channels, and ridge to samples that the previous two do not predict. Taking 0.50 as a clear catch and 0.10 as a miss, ridge clearly catches every fault that SPE clearly catches, and also catches 5 (condenser cooling water inlet temperature), 10 (C feed temperature (stream 4)), 16 (unknown), 20 (unknown), which SPE does not. SPE clearly catches fault 4 and T2 does not (T2 0.091, SPE 0.994, ridge 0.969). No fault is clearly caught by T2 and missed by SPE.

Fault 4 is a step in reactor cooling-water inlet temperature. On run 1, the share of samples above the threshold in samples 21-40, then 200-500, is T2 0.15 then 0.35, SPE 0.90 then 1.00, and ridge 1.00 then 0.99. The temperature controller moves the reactor cooling-water valve and returns the measurements toward the normal region, so T2 alarms only briefly (rate 0.091, although every run alarms at least once). The valve and the reactor temperature no longer follow their normal relationship, so SPE and the ridge residual stay high.

Fault 5 is a step in condenser cooling-water inlet temperature, and the detectors disagree most here. On run 1 all three exceed the threshold just after the fault (T2 0.90, SPE 0.90, ridge 1.00). Late in the run T2 is at 0.03 and SPE at 0.04, while ridge stays at 1.00. The controllers return the operating point inside both PCA limits, but the one-step forecast stays wrong. Over the 20 runs the rates are T2 0.332, SPE 0.125 and ridge 0.996. Fault 7 in Figure 1 is the opposite case: the disturbance is not absorbed and the late shares stay high (T2 1.00, SPE 1.00, ridge 1.00).

5. Faults nobody catches

All three statistics stay below 0.10 on 3 (D feed temperature (stream 2)), 9 (D feed temperature (stream 2)), 15 (condenser cooling water valve), 19 (unknown). Faults 3, 9 and 15 are the faults the project brief and Russell, Chiang and Braatz (2000) single out as hard to detect. Fault 3 is a step in D-feed temperature, fault 9 is random variation in the same temperature, and fault 15 is a sticking condenser cooling-water valve. Rates ($T2 / SPE / \text{ridge}$) are 0.000 / 0.000 / 0.000 for fault 3, 0.000 / 0.000 / 0.000 for fault 9 and 0.000 / 0.000 / 0.000 for fault 15. Runs missed are 19 / 20 / 20, 19 / 20 / 20 and 19 / 20 / 20. The single $T2$ alarm on each is the run-7 excursion described in Section 3.

These faults either change the recorded channels too little or are cancelled by the control loops before they reach them, so the samples stay within the variation of the 300 normal training runs. Missing them reflects the sensors and the controllers, not a healthy plant. Fault 19 is different. Its rates are low, but runs missed are only 0 / 6 / 0, so it is weak rather than invisible.

6. Diagnosis

SPE and the ridge score are sums of squared terms, one per channel. For each fault, those terms are averaged over every sample in alarm after sample 20 across the 20 runs, and the five largest are kept. A contribution shows where a fault appears, not necessarily where it starts: Westerhuis, Gurden and Smilde (2000) show that contributions smear onto correlated channels.

Fault 1: A/C feed ratio, B composition constant (stream 4); step. SPE top five: $x_{\text{meas_4}}$ (A and C feed (stream 4)); $x_{\text{mv_4}}$ (A and C feed flow valve (stream 4)); $x_{\text{meas_38}}$ (product composition E); $x_{\text{meas_21}}$ (reactor cooling water outlet temperature); $x_{\text{meas_33}}$ (purge composition E). Ridge top five: $x_{\text{meas_1}}$ (A feed (stream 1)); $x_{\text{mv_3}}$ (A feed flow valve (stream 1)); $x_{\text{meas_16}}$ (stripper pressure); $x_{\text{meas_19}}$ (stripper steam flow); $x_{\text{meas_7}}$ (reactor pressure). SPE ranks the stream-4 flow and its valve highest, where the fault enters. Ridge ranks the A feed and its valve highest, where the controllers correct the ratio. Product composition E and the reactor cooling-water outlet temperature are downstream effects.

Fault 4: reactor cooling water inlet temperature; step. SPE top five: $x_{\text{mv_10}}$ (reactor cooling water flow valve); $x_{\text{meas_9}}$ (reactor temperature); $x_{\text{meas_33}}$ (purge composition E); $x_{\text{meas_25}}$ (reactor feed composition C); $x_{\text{meas_29}}$ (purge composition A). Ridge top five: $x_{\text{mv_10}}$ (reactor cooling water flow valve); $x_{\text{meas_9}}$ (reactor temperature); $x_{\text{mv_3}}$ (A feed flow valve (stream 1)); $x_{\text{meas_1}}$ (A feed (stream 1)); $x_{\text{meas_26}}$ (reactor feed composition D). The inlet temperature is not measured. Both detectors rank the reactor cooling-water valve first and the reactor temperature second, which is the loop that responds to the disturbance. The remaining SPE terms are small, so the ranking is not smeared.

Fault 6: A feed loss (stream 1); step. SPE top five: $x_{\text{mv_5}}$ (compressor recycle valve); $x_{\text{mv_10}}$ (reactor cooling water flow valve); $x_{\text{meas_9}}$ (reactor temperature); $x_{\text{meas_20}}$ (compressor work); $x_{\text{meas_1}}$ (A feed (stream 1)). Ridge top five: $x_{\text{meas_1}}$ (A feed (stream 1)); $x_{\text{mv_3}}$ (A feed flow valve (stream 1)); $x_{\text{meas_10}}$ (purge rate (stream 9)); $x_{\text{mv_6}}$ (purge valve (stream 9)); $x_{\text{mv_5}}$ (compressor recycle valve). Ridge ranks the A feed and its valve first, the stream that was lost. SPE ranks the compressor recycle valve first, then reactor cooling and compressor work, with the A feed fifth. SPE shows the plant-wide correction: losing the A feed changes the recycle flow and the reactor heat balance, which matches the smearing described by Westerhuis, Gurden and Smilde.

7. Limits

Cut-offs. The 0.10 line for a catch and the 0.50 line for a clear catch are our own choices, not the project's, and they decide which faults Sections 4 and 5 call missed. Fault 19, at 0.029 or less on every statistic, would change category with a lower line. The thresholds are empirical percentiles, not the textbook F and Jackson-Mudholkar limits, which assume independent normal samples.

Alarm rule. At 480 samples a day, raw exceedances on the test runs occur about 5.2, 4.8 and 4.8 times a day for $T2$, SPE and ridge, close to the 4.8 a day reported in Lecture 8 for a 99th-percentile threshold. The three-sample rule cuts alarmed samples to 0.25, 0.00 and 0.00 a day, at the cost of at least 6 extra minutes before any alarm. A real plant would choose this rule by weighing the cost of a false alarm against the cost of a late one.

Sample size. Zero alarmed samples out of 50,000 in 100 test runs shows the false-alarm rate is small, not zero, and correlated neighbouring samples make the evidence weaker than the count suggests. Each fault has only 20 runs, and runs with the same number appear to share background noise across faults, so differences between faults rest on fewer independent draws than the table suggests. Every metric also assumes a known fault start after sample 20. A real plant does not know when a fault began, and operators care about alarm events and time to first alarm more than the share of samples in alarm.

Scope. The data are one simulated operating mode with scripted faults. They contain no drift in normal operation, no failed sensors and no faults during setpoint changes. Contribution ranks are not root causes. The ridge model is linear with two lags, so a slow oscillation can look predictable. A fault that the controllers fully absorb is invisible to all three statistics, whatever the threshold.

8. AI use

Cursor's coding agent (Grok) drafted the pipeline and the first version of this report, constrained to follow the project recipe: runs 1-300 for fitting, 301-400 for thresholds and 401-500 for false alarms, $\text{ddof} = 1$ standardization, a 90% PCA cutoff, Ridge($\alpha = 1$) on lags $t-1$ and $t-2$, 99th-percentile thresholds, the three-sample alarm rule and top-five contributions. Claude Code (Anthropic) was

used to review the evaluation code and to revise the text of this report. Every number in the report is computed from the score files, not typed in, and the PDF was not edited by hand. The team checked the code against the recipe. The course evidence script, run on 8 October 2026, passed all automatic checks (10/10). Each member is responsible for explaining the code they own and the comparison in Section 4.

Appendix: contributions

This appendix does not count toward the four-page limit. One entry per member.

Xinran Li (xinranl3). Part 1, PCA monitor (T2 and SPE). Code: data.py (checksums, ddof = 1 standardization), pca_monitor.py, plots.py.

Mingyao Xu (mingyaox). Part 2, forecast-residual monitor (ridge on lags t-1, t-2). Code: ridge_monitor.py (lag_design, fit_ridge, score_ridge).

Mark Tan (marktan). Part 3, evaluation and submission (thresholds, alarms, detection table). Code: evaluate.py (validation_threshold, alarm_mask, detection_rows), the self-checks in pipeline.py (_check_alarm_rule, _assert_score_rows); ran the course evidence script.

Sean Chao (hsuanlec). Part 4, diagnosis (contributions and the fault list). Code: diagnose.py; FAULT_DESCRIPTION and FAULT_EXPECTED_CHANNELS in config.py.